Homework 7-8, Probability

1) The weight of Scrummy chocolate bars is Normally distributed with a mean of 120 g and standard deviation of 5 g.

Find the probability that the mean weight of a random sample of 16 such chocolate bars exceeds 122 g.

2) A machine is known to dispense liquid into cartons such that the volume dispensed is Normally distributed with a standard deviation of 3.4 ml. It is claimed by the manufacturer of the machine that the mean volume dispensed is 100 ml.

In a series of trials on the machine the following values were recorded:

104.4 101.7 103.6 99.2 102.2 107.6 96.2 105.8 104.7 98.6

a) Find a 95% symmetrical confidence interval for the mean volume of liquid dispensed by the machine.
b) Use your interval to comment on the claim of the manufacturer.
c) Comment on the likely reaction of customers receiving cartons from the machine.

3) The maximum load a lift can carry is 450 kg. The weights of men are Normally distributed with mean 60 kg and standard deviation 10 kg. The weights of women are Normally distributed with mean 55 kg and standard deviation 5 kg. Find the probability that 5 men and 2 women will overload the lift, if their weights are independent.

4) A random sample of 100 Globrite light bulbs had a mean lifetime of 1400 hours with a standard deviation of 150 hours. A random sample of 200 Luxiglo light bulbs had a mean lifetime of 1350 hours and a standard deviation of 140 hours. Stating your hypotheses clearly, test, at the 1% level of significance, whether or not there is a difference between the mean life times of these two types of bulb.

5) The number of accidents per day on a stretch of a motorway was recorded and the following results were obtained over a period of 100 days:

Number of accidents	0	1	2	3	4	5
Number of days	44	32	9	10	5	0

Examine whether or not a Poisson model is suitable to represent the number of accidents per day on this stretch of motorway. Use a 1% level of significance.

6) A survey of the number of hours spent watching television each week was carried out. The sample, randomly selected from a large population, contained 240 males and 200 females. The results are summarised below.

	Male	*Female*
Under 10 hours	40.0%	32.5%
10-25 hours	47.5%	51.5%
Over 25 hours	12.5%	16.0%

Use an approximate χ^2 statistic at the 5% level of significance to assess whether or not there is any association between gender and time spent watching television.

7) Prove that if X and Y are *independent* random variables, than $Var(X+Y) = Var(X) + Var(Y)$, and also $Var(X-Y) = Var(X) + Var(Y)$. (Use the properties of expectation and variance, discussed in lecture 5-6, and also that $E(XY) = E(X)E(Y)$ for independent random variables.)

8) Let X be a random variable Exponentially distributed with parameter λ (for example, the waiting time until a bean we planted will hatch). Let $Y = [X] + 1$ (the integer part of X +1). Prove (using the cumulative distribution function, defined as a probability) that the distribution of Y is Geometric. Find its parameter.

9) Prove the following result for the moment generating function of the Standard Normal distribution:
$M(s) = E(e^{sX}) = \int_{-\infty}^{\infty} e^{sx} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = e^{s^2/2}$.

Hint: Complete the square in the integrand, and use $\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx = 1$, or in more general form, $\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} dx = 1$ (which is the integral of the Normal density function, giving 1, the total probability).

10) The probability of throwing a 6 with a biased die is p . It is known that p is equal to one or other of the numbers A and B where $0 < A < B < 1$. Accordingly the following statistical test of the hypothesis H_0: $p = B$ against the alternative hypothesis H_1: $p = A$ is performed.

The die is thrown repeatedly until a 6 is obtained. Then if X is the total number of throws, H_0 is accepted if $X \leq M$, where M is a given positive integer; otherwise H_1 is accepted. Let α be the probability that H_1 is accepted if H_0 is true, and let β be the probability that H_0 is accepted if H_1 is true.

Show that $\beta = 1 - \alpha^K$, where K is independent of M and is to be determined in terms of A and B. Sketch the graph of β against α.

(This was 14th question in STEP II 2003.)